

JEE MAIN CRASH COURSE

IT'S TIME TO LEARN APPLICATION

- ✓ LATEST NTA PATTERN
- ✓ CONCEPT VIDEOS
- ✓ LIVE DOUBT SESSIONS
- ✓ DPP, REVISION SHEETS
- ✓ 100% MONEYBACK TO MERITORIOUS STUDENTS

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YOUR LAST CHANCE!

Designed by
Math Maestro

Anup Sir



JEE ADV. CRASH COURSE

TARGETING JEE ADVANCED 2020

- ✓ CONCEPT BUILDING VIDEOS
- ✓ ADVANCED LEVEL VIDEOS
- ✓ 2 LEVELS OF SHEETS,
1 WITH ALL TYPE OF QUES.
- ✓ LIVE DISCUSSION OF
PREVIOUS YEAR QUESTIONS

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JEE Mains 2020 Jan Chapter wise Question Bank

Hyperbola

Q1

If a hyperbola has vertices $(\pm 6, 0)$ and $P(10, 16)$ lies on it, then the equation of normal at P is

यदि एक अतिपरवलय के शीर्ष $(\pm 6, 0)$ हैं तथा बिंदु $P(10, 16)$ इस अतिपरवलय पर स्थित है, तब P पर अभिलम्ब का समीकरण है

(1) $2x + 5y = 100$

(2) $2x + 5y = 10$

(3) $2x - 5y = 100$

(4) $5x + 2y = 100$

8th Jan Evening

Sol

(1)

Vertex is at $(\pm 6, 0)$

$$\therefore a = 6$$

Let the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Putting point $P(10, 16)$ on the hyperbola

$$\frac{100}{36} - \frac{256}{b^2} = 1 \quad \Rightarrow \quad b^2 = 144$$

$$\therefore \text{hyperbola is } \frac{x^2}{36} - \frac{y^2}{144} = 1$$

$$\therefore \text{equation of normal is } \frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2$$

$$\therefore \text{putting we get } 2x + 5y = 100$$

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